WORKED SOLUTION

STEP I 2003, Question 8

Pure • Logistic Differential Equation

STEP I 2003, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s1-q8

The key idea

Translate "the rate at which A converts into B" into a derivative: it is the rate at which the volume of B grows. Because $x = 1 - y$, this gives a differential equation in y alone with separable variables, and partial fractions finish the integration. The final part is about whether the formula for $x = 1 - y$ can ever reach zero, and the answer rests on the fact that an exponential is never infinite.

Solution

Setting up the equation

The volume of B is yV, and V is fixed. A turns into B at rate $kVxy$, so the volume of B increases at that rate:

$$\frac{\mathrm{d}}{\mathrm{d}t}(yV) = kVxy \quad \Longrightarrow \quad \frac{\mathrm{d}y}{\mathrm{d}t} = kxy = ky(1-y),$$

dividing by V and using $x = 1 - y$. (Equivalently, the volume of A falls at this rate: $V\,\mathrm{d}x/\mathrm{d}t = -kVxy$, which says the same thing because $\mathrm{d}x/\mathrm{d}t = -\mathrm{d}y/\mathrm{d}t$.)

Solving it

At the start $x > 0$ and $y > 0$, so $0 < y < 1$. While $0 < y < 1$ we can divide by $y(1-y)$ and separate the variables:

$$\int \frac{\mathrm{d}y}{y(1-y)} = \int k\,\mathrm{d}t.$$

By partial fractions, $\dfrac{1}{y(1-y)} = \dfrac{1}{y} + \dfrac{1}{1-y}$, so

$$\ln y - \ln(1-y) = kt + c,$$

$$\ln \frac{y}{1-y} = kt + c,$$

where we've used that y and $1-y$ are both positive, so no modulus signs are needed. Exponentiating and writing $D = \mathrm{e}^c$,

$$\frac{y}{1-y} = D\mathrm{e}^{kt}.$$

Then $y = D\mathrm{e}^{kt} - yD\mathrm{e}^{kt}$, so $y\left(1 + D\mathrm{e}^{kt}\right) = D\mathrm{e}^{kt}$ and

$$y = \frac{D\mathrm{e}^{kt}}{1 + D\mathrm{e}^{kt}},$$

as required. Putting $t = 0$ shows $D = \dfrac{y_0}{1-y_0} = \dfrac{y_0}{x_0}$, where x_0 and y_0 are the starting values; in particular $D > 0$.

Ask yourself We assumed $0 < y < 1$ to do the integration. How do we know this stays true for all time?

Suppose not, and let T be the first time at which y reaches 0 or 1. On $0 \le t < T$ the working above is valid, so y is given by the formula there. Since y is continuous, $y(T)$ is the limit of the formula as $t \to T$, namely $D\mathrm{e}^{kT}/(1 + D\mathrm{e}^{kT})$. With $D > 0$ this lies strictly between 0 and 1, which is a contradiction. So $0 < y < 1$ for all t, and the formula holds for all t.

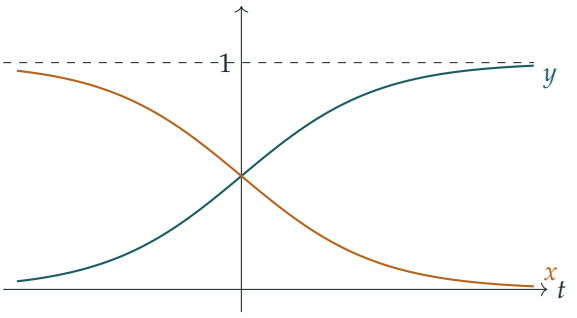
Does A ever completely convert?

No. From the formula,

$$x = 1 - y = \frac{1}{1 + D\mathrm{e}^{kt}},$$

and since $D\mathrm{e}^{kt}$ is finite for every t, $x > 0$ at every time: some of A is always left. However, as $t \to \infty$, $D\mathrm{e}^{kt} \to \infty$ (because $k > 0$ and $D > 0$), so $x \to 0$ and $y \to 1$. The conversion gets ever closer to complete but never finishes in finite time.

This makes physical sense: the reaction rate $kVxy$ is proportional to the amount of A left, so the reaction slows down as A runs out.



The sketch (with $D = k = 1$) shows the S-shaped logistic curve for y approaching 1 and the curve for x approaching 0, neither ever reaching its limit.

Checks

- Differentiate the answer: with $u = De^{kt}$, $y = u/(1+u)$ and $\mathrm{d}y/\mathrm{d}t = ku/(1+u)^2 = ky(1-y)$, as it should.
- With $D = k = 1$ (so we start with equal volumes) the formula gives $y = \frac{3}{4}$ at $t = \ln 3$. We also stepped the equation forward numerically for 200 random choices of k and starting value, and it matched the formula to within 10^{-12}, with x staying positive throughout.

Common slips

- Treating "$y = 1$ in the limit" as "$y = 1$ at some time". A limit as $t \to \infty$ is not attained at any finite time.
- Forgetting that the step $\ln(y/(1-y))$ needs $0 < y < 1$, or not saying why that condition persists.

Beyond STEP

The equation $\dot{y} = ky(1-y)$ is the logistic equation, the standard model for growth that is limited by a resource. Here $y = 0$ and $y = 1$ are equilibrium solutions, and the uniqueness theorem for differential equations says no other solution can ever cross or touch them, which is a slicker way to settle the final part. Boyce and DiPrima, *Elementary Differential Equations*, discusses the logistic equation and this kind of stability analysis in detail.

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